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Editor-in-Chief, Senior Editor, and Associate Editor *Information Systems Research*

Re: Submission of a Research Article, "Who Pays for the Next Attempt? Contract Design and Escalation Rules for AI-Assisted Software Maintenance"

Dear Editors,

We submit the enclosed manuscript for consideration as a Research Article in *Information Systems Research*. The paper is a theory article that applies and extends search theory and vertical-contracting theory to a new information systems phenomenon, pay-per-use AI services, and illustrates the model with open repair data. It addresses a question that is becoming material for IT organizations: when a vendor of AI-based software repair charges per attempt or per verified fix, how many failed attempts should the buyer tolerate before escalating a defect to an engineer, and how does the vendor's price structure shape that decision? We derive the buyer's optimal escalation rule, characterize the vendor's optimal contract and the rent that deep sampling requires the vendor to leave the buyer, and examine private vendor quality and competition. We test the premises of the model on open data and calibrate its cost parameters to public wage statistics, task-time annotations, and API price lists.

The intended audience is both academic (researchers on the economics of information systems, IT sourcing and governance, and human–AI work design) and practical (chief information officers, engineering and procurement managers, and AI vendors).

Contribution statement

(a) Contribution to the literature. Research on AI–human collaboration in information systems studies how people delegate to and learn from AI, and research on IT service contracting studies how fixed-price, time-and-materials and outcome-based contracts allocate risk between client and vendor. Generative AI services create a setting that neither literature addresses directly: output is stochastic and checkable, the same task can be attempted repeatedly, and the vendor can charge per attempt or per verified result. We model this as a vendor–buyer game in which the buyer chooses how many failed attempts to tolerate before escalating to a human, and defects differ in how easily they are repaired, so that every failure is informative about the next attempt.

The paper makes four contributions. First, it shows that the buyer's optimal escalation rule is a reservation threshold that holds for any distribution of repair difficulty, and that a homogeneous model, which ignores the information in failures, would prescribe never stopping. Second, it identifies an *incentive rent of sequential search*: because later attempts are less likely to succeed than earlier ones, any pay-per-use

contract that induces deep sampling must leave the buyer a gain on the earlier attempts. This rent explains why a per-attempt price above cost makes buyers stop too early (double marginalization, applied to attempts), why a monopolist under-supplies sampling even under outcome-based pricing, why a pure success fee is the vendor's best pay-per-use contract, and why a lump-sum fee would remove all distortions. Third, it shows that private vendor quality reverses the profit order across contract forms, so that only outcome-based pricing can credibly signal quality, and that competition removes the distortion only when vendors are close substitutes, not merely numerous. Fourth, it tests the model's premises on open repair data (574 bugs, 45 models, plus a new 40-sample experiment) and calibrates costs from public wage statistics, task-time annotations and API price lists, showing where the conclusions hold (when verification is cheap relative to human repair) and where they weaken.

(b) Influence on practice, policy and society. Organizations are signing contracts for AI-assisted maintenance without a way to choose how long the AI should keep trying or how to read a vendor's price structure. The paper gives buyers an operational rule (estimate the dispersion of repair difficulty from their own logs, compute the threshold from their cost of human repair and the full cost of an attempt, and validate it on held-out defects), and it shows that pass@k benchmarks measure capability, not the economic decision. It gives vendors and buyers a basis for choosing between per-attempt and outcome-based pricing and for anticipating how market structure shifts the choice. For policy and standards bodies, it shows that the inputs the contracts depend on, per-sample outcomes and per-attempt costs, are not disclosed in current public leaderboards, which supports disclosure requirements in AI procurement. The analysis also helps interpret how AI services may shift routine maintenance work between machines and engineers, and who captures the resulting savings.

Paper provenance

[Authors to confirm each item before submission.] The manuscript has not been published and is not under review elsewhere. It has not previously been submitted to *Information Systems Research*. [State whether any version has been submitted to another journal and, if so, the reasons for rejection and the changes made.] [State whether the paper is posted on a working-paper website other than the authors' own, and whether any part has been copyrighted for pre-publication.] [List any related work by the authors that overlaps with this submission.]

Data provenance

The manuscript uses only public data and public information. We list the sources and the status of the permissions below.

- *RepairBench results* (public GitHub repository of the RepairBench project, retrieved at the commit of 26 October 2025). We use counts of test-passing samples and token usage; no patches are redistributed. The repository carries no license file, so we have asked or will ask the maintainers for written permission. [Status of permission: pending; attach documentation if required.]
- *SWE-bench submissions* (public GitHub repository of the SWE-bench experiments). We use resolved lists and statements of pass@k and cost in submission READMEs. The checkout we used had no license file at its root. [Status of permission: pending.]
- *HumanEvalPack* (HumanEvalFix, Python; Hugging Face dataset, MIT license). We use the 164 tasks and their tests. We generated 19,680 model samples with three open-weight models through a public API and store only pass/fail outcomes and token counts, not generated code. [Authors to confirm that the terms of the API and of the three model licenses permit this use.]
- *Price list and statistics.* Public model prices from the OpenRouter model list (accessed 25 October 2026); wage and compensation statistics from the US Bureau of Labor Statistics; task-time annotations for SWE-bench Verified as summarized by Epoch AI. No human-subject data are collected or used.

Use of generative AI

[Authors to complete in accordance with the INFORMS policy on generative AI. The manuscript's empirical component itself uses language models as an object of study: three open-weight models were sampled to produce the repair outcomes analyzed in Section 6. Describe any use of AI tools in drafting text, writing code, or checking proofs, and confirm that the authors reviewed and take responsibility for all content.]

Preferred editors and reviewers

[Two preferred Senior Editors; two preferred Associate Editors; five preferred reviewers, at least two of them members of the Editorial Review Board; none with a conflict of interest as defined in the submission guidelines.]

Funding and conflicts of interest

[Funding statement.] [Conflict-of-interest statement.]

Thank you for considering our work. We would be glad to provide any additional information.

Sincerely,

[Corresponding author name] [Affiliation and e-mail]